

multivariate (A/B) test Analysis:

Period: March 3–18

Buckets: 3–10 & 11–18

Variants: 0, 1, 2

KPIs definitions & scope

- Daily new users (DNU) / New users (NU): Number of unique users opened the app for the first time in the day / period.
- Daily Active users (DAU) / Active users(AU) :Number of unique users active in the day / period.
- Paying users (PU): Number of unique users who made at least one purchase in the day / period.
- Daily revenue : Sum of in-app purchases + ad-watched revenue.
- Average revenue per daily active user (ARPDAU): total revenue / DAU
- Average revenue per paying user (ARPPU): in-app purchases revenue / PU

KPIs definitions & scope

- Retention Rate % : percentage of prior period users who returned during the experiment period.

Retention Rate % = (Users at the end of period – NU)/ Users at the start of period * 100

- D9 Retention is measured (March 11) against Day16 (March 18), with Day0 as March 3, detailed calculations are provided later in the slides.

- Churn Rate % : percentage of prior period users who returned during the experiment period.

Churn Rate % = 1 – retention

- Life Time of a user (in days) (LT) : period expected for a user to keep using the app until churned.

LT = (1/ churn rate) * # of days of the experiment.

- Lifetime value (in days) (LTV): expected cumulative revenue per user over their lifetime

LTV = ARPDAU * LT

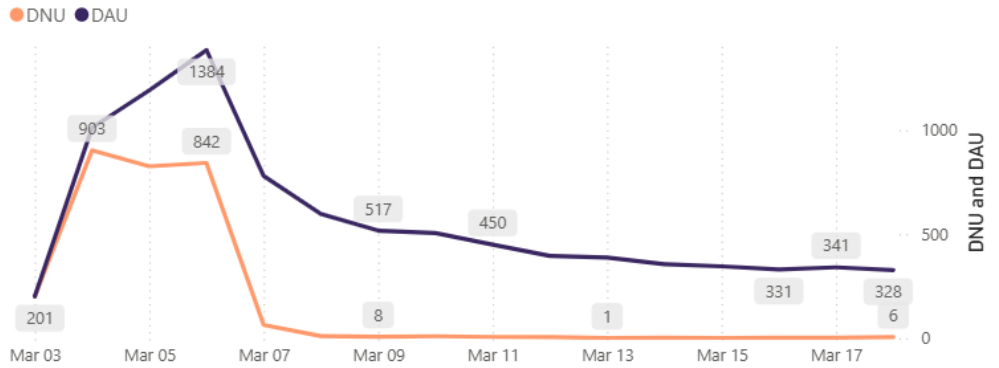
Disclaimer:

- LTV, LT, Retention Rate, and Churn Rate are calculated based on data from March 11–18, representing half of the available period (March 3–18).
- This was done solely for the purpose of this task, as these KPIs require longer historical data for accurate evaluation. Commonly 28, 60, 90 days in many mobile games.
- Results may therefore be indicative.

Engagement & Monetization: DAU & DNU

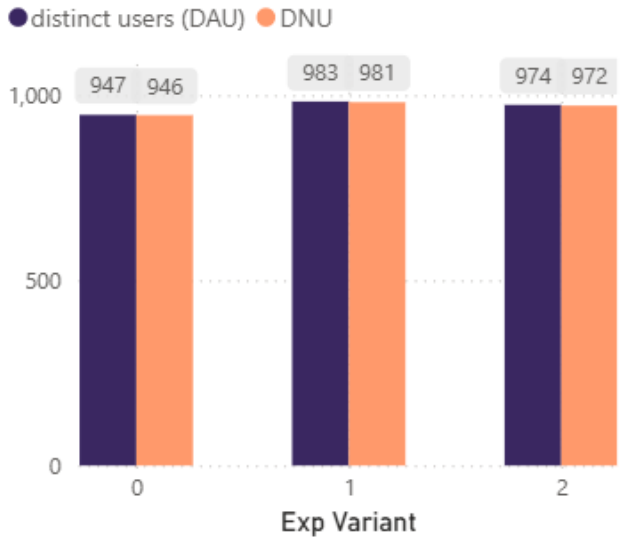
Observed trends (Mar 3–18)

- **Sharp spike early March** peaking with (**1384 users**) on Mar 6, then dropping to a steady baseline **(~300–400**) for bucket 2 (Mar 11th -18th).
- Daily new users are small during bucket2 (**31 total**), so DAU is sustained mostly by returning users (no new installs during the second week).



Behavioral interpretation

- The early spike likely reflects a marketing push or campaign driving many installs for a short period.
- The post-spike baseline suggests the campaign either ended or those new users had low retention.
- Low daily new users in bucket 2 means organic acquisition is limited or paused.
- Growth is not sustainable without marketing activities.

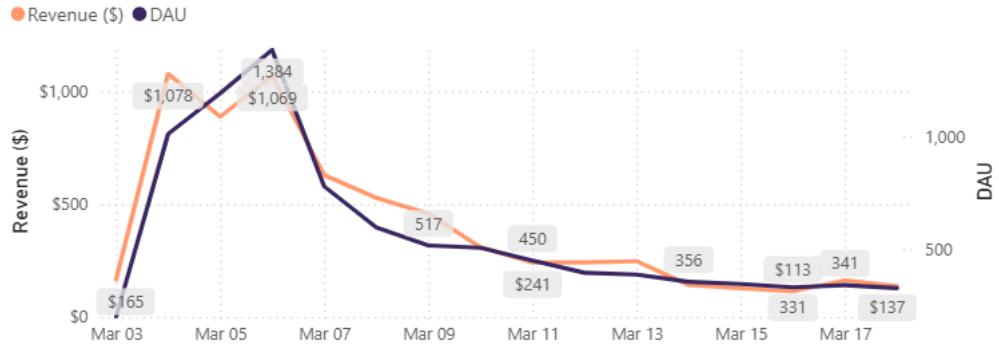


Variant-level comparison

- Variant exposure roughly balanced across variants with:
  - V0 → **947**
  - V1 → **983**
  - V2 → **974**
- No variant massively changed DAU within the 16-day window.

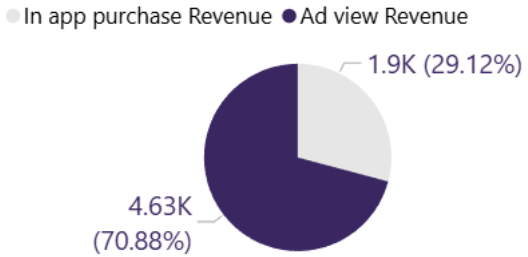
Engagement & Monetization: DAU & Revenue $

Observed trends Mar 3–18)
DAU drop correlates with revenue drop, Revenue concentrated on Mar 4–6 **(> $1,000)** while during bucket2 (**total = $1,410).**

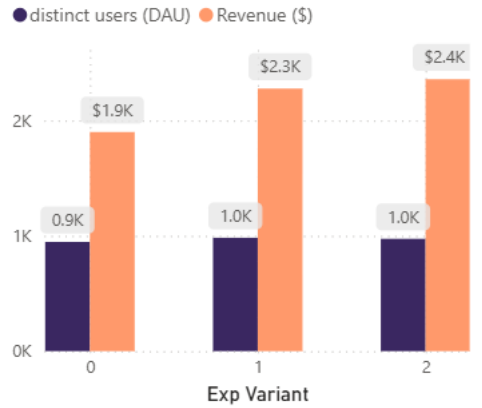


Behavioral interpretation

- **Monetization appears volume-sensitive**, not exclusively dependent on per-user spend mainly because Revenue depends heavily on **ad revenue;** impressions.
- This is generally positive, as it means our revenue base is diversified and less risky and not concentrated in a few “whales”.
- However, the downside is that when retention fails, overall revenue suffers significantly. Since each active user contributes modestly, maintaining a large, engaged player base is critical.

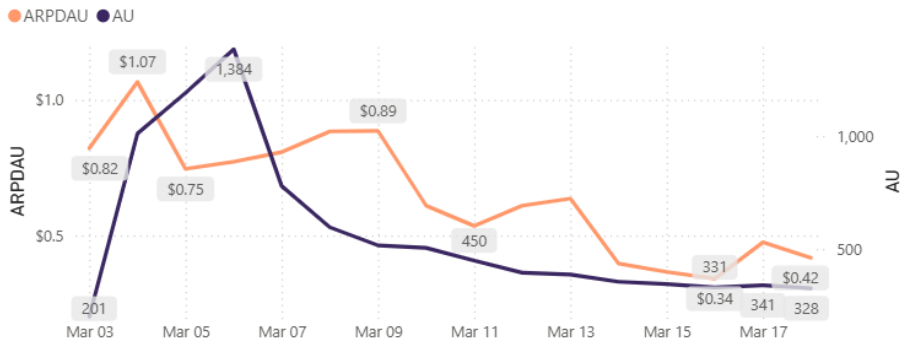


Variant-level comparison
Variants 1 & 2 outperform V0 in overall revenue:
V2 highest, likely due to more total players.
V1 nearly equal but from fewer but higher-value payers.

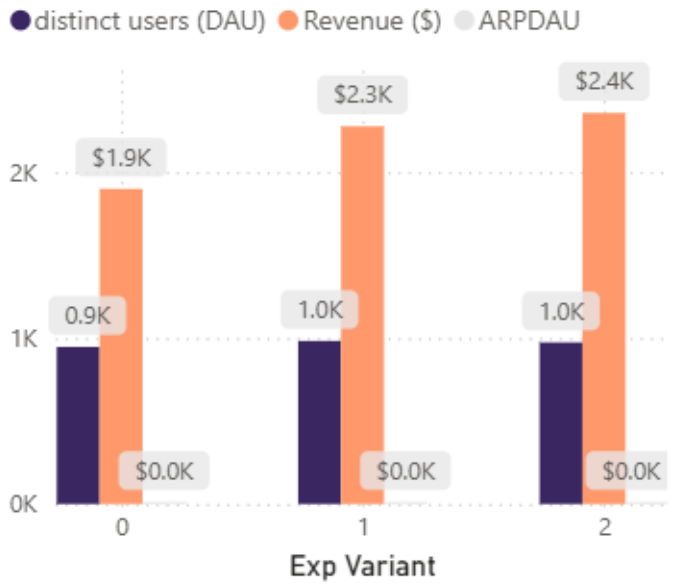


Engagement & Monetization: DAU & ARPDAU

Observed trends
- **ARPDAU peaks with revenue spikes early** (4-6 March).
- In bucket2 ARPDAU is lower ($0.4 - $0.6 daily).



Behavioral interpretation
- Since **70% of revenue is ad-based**, this drop isn't about paying users leave, it's about **engagement quality** among the retained players.
- As users stay longer, they open the app less frequently, spend shorter sessions, and watch fewer ads per session; **engagement fatigue.**



Variant-level comparison
- V2 → Most Efficient Ad Monetization:
  - Highest ARPDAU ($2.42)
  - Highest total revenue ($2,357.84)
- Since paying users are nearly identical, the difference almost certainly comes from more ad views per active user:
  - Higher session frequency
  - Better ad exposure mechanics.

Engagement & Monetization: PU & In app-purchase Revenue

Observed trends Mar 3–18)

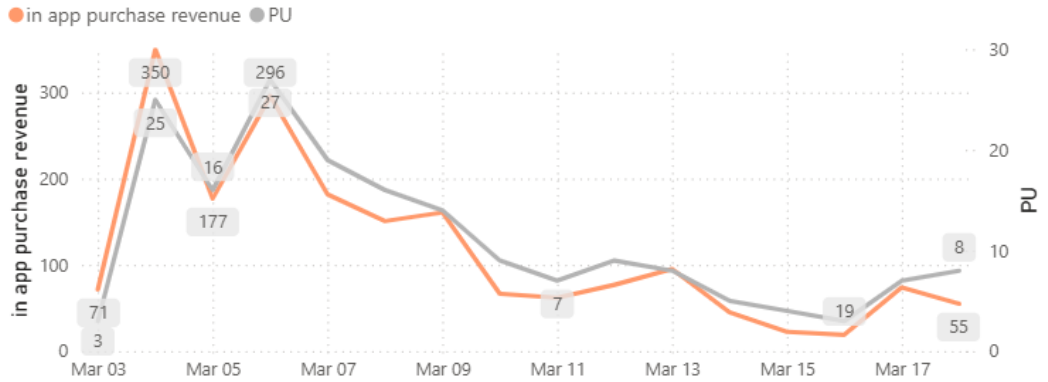
- Paying users concentrated in spikes and small across bucket2 **(33 PU**).

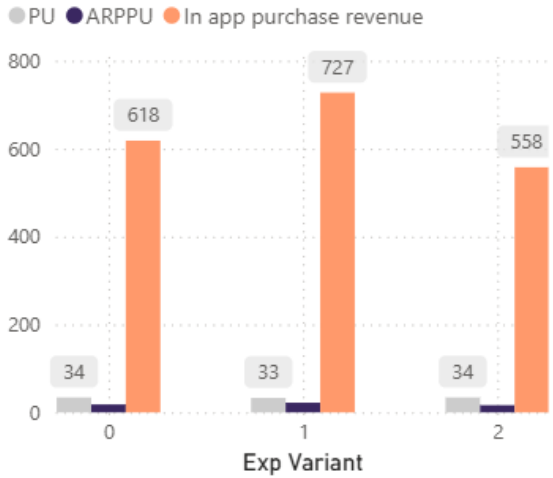
Behavioral interpretation

- **Only few players actually spend money**, so revenue (in app-purchase revenue) changes a lot when even a few of those players spend more or less.
- If those few paying users happen to spend on some days and not others, revenue graph will swing a lot.
- Low conversion rate causes low revenue even if DAU is moderate.
- Focus on increasing **payer conversion rate** rather than only increasing DAU.

Variant-level comparison

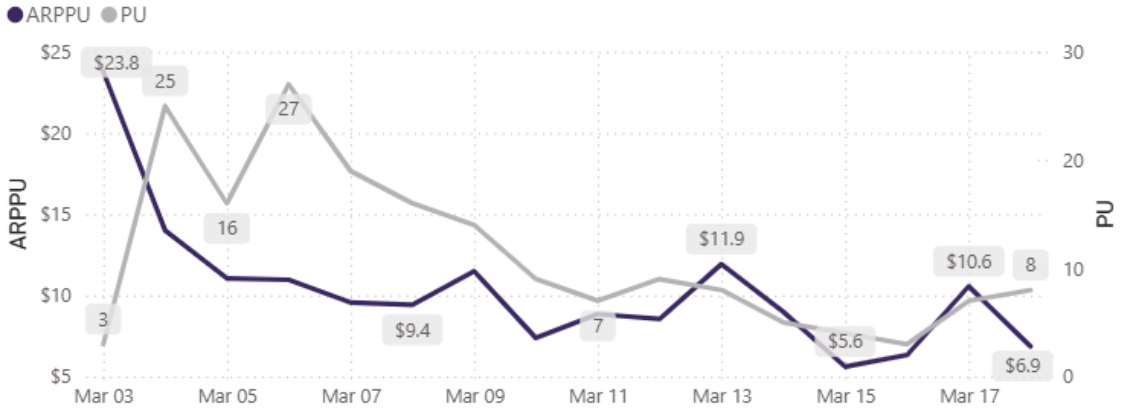
- V1 is performing better in terms of ARPPU and revenue.
- However, Since the group of paying users is **so small**, it's hard to tell whether V1 is really performing better or if the difference is just **random chance.**





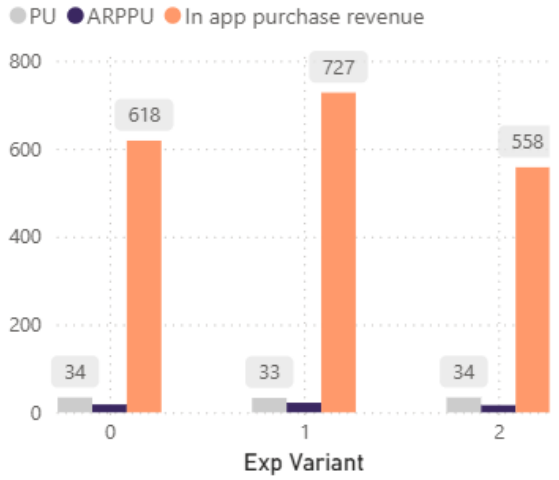
Engagement & Monetization: PU & ARPPU

Observed trends
ARPPU varies widely day-to-day due to small paying user counts.
However, it generally follows the usual trend (high spike then decline)



Behavioral interpretation

- On days with **fewer payers**, ARPPU is **higher**
  - Mar 3rd : 3 payers, $23.80 ARPPU.
- On days with **many payers**, ARPPU **drops**
  - Mar 6th : 27 payers, $10.96 ARPPU.
- This pattern shows that revenue is spread across small groups of inconsistent buyers, not steady mid-tier spenders
- One or two big purchases push up ARPPU, but on high-volume days, more users make smaller purchases.
- This is not catastrophic as total revenue stability relies more on ad engagement.
- But it still matters, IAP spenders are high-intent users, often linked to stronger retention and engagement.



Variant-level comparison

- V1 has substantially higher ARPPU and revenue, indicates the variant's mechanics encourage higher individual spend but unsustainable engagement.

Retention: Retention & Churn rates.

Observed trends (Bucket2 only)
- Day-9 (Mar 11th): **15.46%**.
- Day-16 (Mar 18th): **11.14%**.
- Decline: −4.32 percentage points.
- Healthy range: **Day-14 retention→ 15–25%**

Behavioral interpretation
- Rapid decay indicates engagement or content depth issues; most users disengage before mid-term milestones.
- New users entering during Mar 11–18 are fewer (DNU = 31 total for 8 days) and not compensating for churn.

Variant-level comparison (Bucket2)
- V0 → 10.26%
- V1 → 8.59%
- V2 → 10.58%
- V1 shows the lowest retention but highest immediate monetization; typical monetization (onboarding trade-off).
- V2 retains better, suggesting superior player experience.

event_date	DAU	DNU	Users Before this Period	Retained Users	Retention Rate (%)	Churned Users	Churn rate (%)
Tuesday, March 11, 2025	450	7	2,865	443	15.5%	2,422	84.5%
Wednesday, March 12, 2025	396	6	2,872	390	13.6%	2,482	86.4%
Thursday, March 13, 2025	388	1	2,878	387	13.4%	2,491	86.6%
Friday, March 14, 2025	356	3	2,879	353	12.3%	2,526	87.7%
Saturday, March 15, 2025	346	2	2,882	344	11.9%	2,538	88.1%
Sunday, March 16, 2025	331	3	2,884	328	11.4%	2,556	88.6%
Monday, March 17, 2025	341	3	2,887	338	11.7%	2,549	88.3%
Tuesday, March 18, 2025	328	6	2,890	322	11.1%	2,568	88.9%
Total	**872**	**31**	**2,865**	**841**	**29.4%**	**2,024**	**70.6%**

exp_variant	DAU	DNU	Users Before this Period	Retained Users	Retention Rate (%)	Churned Users	Churn rate (%)
0	305	11	2,865	294	10.3%	2,571	89.7%
1	257	11	2,865	246	8.6%	2,619	91.4%
2	313	10	2,865	303	10.6%	2,562	89.4%
Total	**872**	**31**	**2,865**	**841**	**29.4%**	**2,024**	**70.6%**

Retention: LT & LTV.

Disclaimer (repeated) :Do not trust LTV/LT from an 8-day window.
LTV uses a lifetime estimate derived from the same short dataset which amplifies error.

Observed trends (Bucket2 only)
LTV: **$18.32**
LT ~ 11 days
Variant LTVs: V0 $9.99, V1 $18.68, V2 $14.88.
Variant LTs: (V0, V1, V2) ~9.

$18.32
LTV

11
Life Time of a user (days)

Variant-level comparison
- V1 shows highest LTV currently due to high ARPPU, but it has worse retention.
- If retention continues to be low, true long-term LTV may be lower.

exp_variant	DAU	DNU	Users Before this Period	Retained Users	Retention Rate (%)	Churned Users	Churn rate (%)	LTV	Life Time of a user
0	305	11	2,865	294	10.3%	2,571	89.7%	$9.99	8.9
1	257	11	2,865	246	8.6%	2,619	91.4%	$18.68	8.8
2	313	10	2,865	303	10.6%	2,562	89.4%	$14.88	8.9
Total	**872**	**31**	**2,865**	**841**	**29.4%**	**2,024**	**70.6%**	**$18.32**	**11.3**

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A/B test variants Summary

- Variants 1 and 2 both outperform the control overall.
- Variant 1 drives higher revenue & ARPPU with lower retention.
- While Variant 2 shows stronger ad monetization and slightly better retention.
- Given app reliance on ad revenue (~70%), **Variant 2's performance is more scalable and sustainable.**
- However, all variants share the same retention weakness, players drop off sharply after the first week, signaling content or engagement fatigue.

exp_variant	DAU	ARPDAU	ARPPU	Retention Rate (%)	LTV
0	305	$1.12	$11.3	10.3%	$9.99
1	257	$2.13	$17.1	8.6%	$18.68
2	313	$1.66	$11.9	10.6%	$14.88
Total	**872**	**$1.62**	**$13.6**	**29.4%**	**$18.32**

Summary

- After **the initial acquisition burst** (Mar 3-6), the game quickly entered a retention bottleneck.
- By Day 9 (Mar 11), **only 15% of the original users were active**; by Day 16 (Mar 18), that number fell below 11%. **Retention decay is faster than typical benchmarks**.
- This retention decline mirrored an equivalent fall in revenue per active user, showing that engagement quality and monetization are tightly coupled (**churn is a revenue risk**).
- The ongoing A/B test variants reveal that **V1 boosts short-term spending** at the expense of retention, while **V2 offers a better long-term balance**.
- The second 8-day bucket (Mar 11–18) **is too short for lifetime metrics** like LTV & churn.
- The test appears to start after a marketing spike, meaning users in Mar 11–18 are less organic or more residual. That can **bias variant comparisons**.

What to do next?

Retention:

- Add **mid-term goals** (missions, daily streaks) around Day 5–10 to maintain engagement.
- Introduce **“come back” rewards** (daily login multipliers).
- Delay **monetization prompts** (IAP popups) until after user complete key milestones.

Revenue management:

- Introduce **low-price “starter packs”** ($0.99–$1.99) to convert early users.
- Add **“ad removal” bundle** as a monetization option.

Ad management:

- Introduce **rewarded ads** for key in-game boosts (double rewards, instant refills) ,but don’t increase much **to avoid session fatigue** (limit to 3 - 4 per session)

Data management:

Extend test window to **at least 4–6 weeks**.

